

SQL Cheat Sheet: Views, Stored Procedures and Transactions

Topic	Syntax	Description	Example
Create View	<pre>CREATE VIEW view_name AS SELECT column1, column2, ... FROM table_name WHERE condition;</pre>	A CREATE VIEW is an alternative way of representing data that exists in one or more tables.	<pre>CREATE VIEW EMPLOYEE AS SELECT EMP_ID, F_NAME, L_NAME, B_DATE, SEX, SALARY FROM EMPLOYEES;</pre>
Update a View	<pre>UPDATE OR REPLACE VIEW view_name AS SELECT column1, column2, ... FROM table_name WHERE condition;</pre>	The CREATE OR REPLACE VIEW command updates a view.	<pre>CREATE OR REPLACE VIEW EMPLOYEE AS SELECT EMP_ID, F_NAME, L_NAME, B_DATE, SEX, JOB_TITLE, REG_SALARY, RAW_SALARY FROM EMPLOYEES; 2000 WHERE EMPLOYEES.EMP_ID = 2000 AND EMPLOYEES.</pre>
Drop a View	<pre>DROP VIEW view_name;</pre>	Use the DROP VIEW statement to remove a view from the database.	<pre>DROP VIEW EMPLOYEE;</pre>

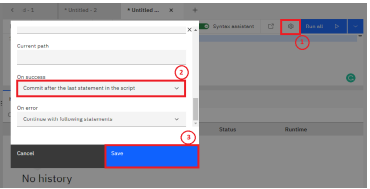
Stored Procedures in IBM Db2 using SQL

Stored Procedures	<pre>--SET TERMINATOR @ CREATE PROCEDURE PROCEDURE_NAME LANGUAGE SQL BEGIN END @</pre>	A stored procedure is a prepared SQL code that you can save, so the code can be reused over and over again. The default terminator for a stored procedure is semicolon (;). To set a different terminator we use SET TERMINATOR clause followed by the terminator such as '@'.	<pre>--SET TERMINATOR @ CREATE PROCEDURE RETRIEVE_ALL LANGUAGE SQL BEGIN SQL DATA DECLARE RESULT SETS 1 DECLARE C1 CURSOR WITH RETURN FOR SELECT * FROM PETSCALE; OPEN C1; END @</pre>
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Stored Procedures in MySQL using phpMyAdmin

Stored Procedures	<pre>DELIMITER // CREATE PROCEDURE PROCEDURE_NAME BEGIN END // DELIMITER ;</pre>	A stored procedure is a prepared SQL code that you can save, so the code can be reused over and over again. The default terminator for a stored procedure is semicolon (;). To set a different terminator we use DELIMITER clause followed by the terminator such as '//'.	<pre>DELIMITER // CREATE PROCEDURE RETRIEVE_ALL() BEGIN SELECT * FROM PETSCALE; END // DELIMITER ;</pre>
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Transactions with Db2

Commit command	<pre>COMMIT;</pre>	A COMMIT command is used to persist the changes in the database. The default terminator for a COMMIT command is semicolon (;).	<pre>CREATE TABLE employees(EMP_ID INT, Name VARCHAR(20), City VARCHAR(20), Salary INT, Age INT); INSERT INTO employees (ID, Name, City, Salary, Age) VALUES (1, 'Priyanka pat', 'Mumbai', 30000, 21), (2, 'Riya chaudhary', 'Bangalore', 32000, 20); SELECT *FROM employees; COMMIT;</pre>
Rollback command	<pre>ROLLBACK;</pre>	A ROLLBACK command is used to rollback the transactions which are not saved in the database. The default terminator for a ROLLBACK command is semicolon (;).	As auto-commit is enabled by default, all transactions will be committed. We need to disable this option to use the rollback works. For Db2, we have to disable auto-commit manually. Click the gear icon located on the right side of the SQL Assistant window. Next, select the "No Submit" from Drop and choose "Commit" after the last statement in the script? Remember to save your changes!  INSERT INTO employees VALUES (3, 'Santha Thiruv', 'Mangalore', 30000, 20); ROLLBACK SELECT *FROM employees; SELECT *FROM employees;

Transactions with MySQL

Commit command	<pre>COMMIT;</pre>	A COMMIT command is used to persist the changes in the database. The default terminator for a COMMIT command is semicolon (;).	<pre>CREATE TABLE employees(EMP_ID INT, Name VARCHAR(20), City VARCHAR(20), Salary INT, Age INT); START TRANSACTION; INSERT INTO employees (ID, Name, City, Salary, Age) VALUES (1, 'Priyanka pat', 'Mumbai', 30000, 21), (2, 'Riya chaudhary', 'Bangalore', 32000, 20); SELECT *FROM employees; COMMIT;</pre>
Rollback command	<pre>ROLLBACK;</pre>	A ROLLBACK command is used to rollback the transactions which are not saved in the database. The default terminator for a ROLLBACK command is semicolon (;).	As auto-commit is enabled by default, all transactions will be committed. We need to disable this option to use the rollback works. For MySQL, use the command "SET autocommit = 0;" INSERT INTO employees VALUES (3, 'Santha Thiruv', 'Mangalore', 30000, 20); SELECT *FROM employees; ROLLBACK SELECT *FROM employees;

Db2 Transactions using Stored Procedure

Commit command	<pre>--SET TERMINATOR @ CREATE PROCEDURE PROCEDURE_NAME BEGIN COMMIT; END @</pre>	A COMMIT command is used to persist the changes in the database. The default terminator for a COMMIT command is semicolon (;).	<pre>--SET TERMINATOR @ CREATE PROCEDURE TRANSACTION_SAVE LANGUAGE SQL MODIFIES SQL DATA BEGIN DECLARE SQLCODE INTEGER DEFAULT 0; DECLARE RETURNING INTEGER 0; DECLARE CONTINUE HANDLER FOR SQLSTATE* SET returncode = SQLCODE; UPDATE BANKACCOUNTS SET Balance = Balance-200 WHERE AccountName = 'Ravi'; UPDATE BANKACCOUNTS SET Balance = Balance-200 WHERE AccountName = 'Ravi'; IF returncode = 0 THEN ROLLBACK WORK; ELSE COMMIT WORK; END IF; END @</pre>
Rollback command	<pre>--SET TERMINATOR @ CREATE PROCEDURE PROCEDURE_NAME BEGIN ROLLBACK; COMMIT; END @</pre>	A ROLLBACK command is used to rollback the transactions which are not saved in the database. The default terminator for a ROLLBACK command is semicolon (;).	<pre>--SET TERMINATOR @ CREATE PROCEDURE TRANSACTION_SAVE LANGUAGE SQL MODIFIES SQL DATA BEGIN DECLARE SQLCODE INTEGER DEFAULT 0; DECLARE RETURNING INTEGER 0; DECLARE CONTINUE HANDLER FOR SQLSTATE* SET returncode = SQLCODE; UPDATE BANKACCOUNTS SET Balance = Balance-200 WHERE AccountName = 'Ravi'; UPDATE BANKACCOUNTS SET Balance = Balance-200 WHERE AccountName = 'Ravi'; IF returncode = 0 THEN ROLLBACK WORK; ELSE COMMIT WORK; END IF; END @</pre>

MySQL Transactions using Stored Procedure

Commit command	<pre>DELIMITER // CREATE PROCEDURE PROCEDURE_NAME BEGIN COMMIT; END // DELIMITER ;</pre>	A COMMIT command is used to persist the changes in the database. The default terminator for a COMMIT command is semicolon (;).	<pre>DELIMITER // CREATE PROCEDURE TRANSACTION_SAVE() BEGIN DECLARE EXIT HANDLER FOR SQLSTATE* ROLLBACK; END; START TRANSACTION; UPDATE BANKACCOUNTS SET Balance = Balance-200 WHERE AccountName = 'Ravi'; UPDATE BANKACCOUNTS SET Balance = Balance-200 WHERE AccountName = 'Ravi'; COMMIT; END //</pre>
Rollback command	<pre>DELIMITER // CREATE PROCEDURE PROCEDURE_NAME BEGIN ROLLBACK; COMMIT; END // DELIMITER ;</pre>	A ROLLBACK command is used to rollback the transactions which are not saved in the database. The default terminator for a ROLLBACK command is semicolon (;).	<pre>DELIMITER // CREATE PROCEDURE TRANSACTION_SAVE() BEGIN DECLARE EXIT HANDLER FOR SQLSTATE* ROLLBACK; END; START TRANSACTION; UPDATE BANKACCOUNTS SET Balance = Balance-200 WHERE AccountName = 'Ravi'; UPDATE BANKACCOUNTS SET Balance = Balance-200 WHERE AccountName = 'Ravi'; COMMIT; END //</pre>

Author(s)

